



Marine microsporidia: Hidden diversity, aquatic emergence and zoonotic risks

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ABSTRACT

Marine Microsporidia represent an increasingly important but historically underestimated group of aquatic intracellular parasites. Although traditionally associated with invertebrates and fishes, recent molecular and environmental studies have revealed extensive hidden diversity across marine ecosystems together with unexpected phylogenetic links between aquatic and human-associated lineages. This review synthesizes five decades of research (1975–2025) concerning the biology, evolution, taxonomy, host–pathogen interactions, ecological diversity, and zoonotic significance of marine Microsporidia. Particular emphasis is placed on members of the Enterocytozoonidae, including *Enterocytozoon hepatopenaei*, *Nucleospora* spp., and *Desmozoon lepeophtherii*, which represent major threats to aquaculture sustainability and may contribute to cross-environmental transmission pathways. Environmental DNA studies further indicate that marine microsporidian diversity is substantially underestimated by morphology-based approaches. However, current evidence for zoonotic transmission remains largely indirect, resting chiefly on phylogenetic relatedness, shared genotypes across hosts and environmental (eDNA) detection; direct, seafood-borne human infection has not yet been demonstrated. Current evidence suggests that aquaculture intensification, climate-driven environmental change, and expanding seafood trade may facilitate the emergence and dissemination of zoonotic microsporidian lineages. Integrating marine Microsporidia into One Health surveillance frameworks is therefore increasingly important for both ecosystem and public health management.

1. Introduction

Microsporidia are obligate intracellular eukaryotic parasites characterized by environmentally resistant chitinous spores and a highly specialized invasion apparatus known as the polar tube. During germination, the polar tube rapidly extrudes and injects the infective sporoplasm directly into the host-cell cytoplasm, enabling intracellular establishment and replication [1–4].

Historically, Microsporidia were regarded as primitive protozoa or “amitochondrial archezoa”. However, molecular phylogenetic and phylogenomic studies have firmly demonstrated that they represent highly reduced fungal-related parasites that evolved through extensive genome reduction and metabolic dependence on host cells [5–7].

More than 1500 described microsporidian species infect an exceptionally broad host spectrum including protists, annelids, crustaceans, molluscs, fishes, amphibians, birds, mammals and humans [8,9]. Despite the medical importance of genera such as *Enterocytozoon* and

Encephalitozoon, aquatic ecosystems constitute the principal reservoir of microsporidian diversity [10,11]. Marine habitats, in particular, harbor phylogenetically diverse and ecologically specialized lineages infecting fishes, crustaceans, copepods and bivalves.

The growing application of molecular phylogenetics and environmental DNA metabarcoding has profoundly changed current understanding of marine Microsporidia. Classical morphology-based taxonomy underestimated diversity because many species exhibit convergent spore morphologies, cryptic developmental stages and limited diagnostic ultrastructural characters [12,13]. Environmental sequencing studies now reveal that a substantial proportion of marine microsporidian operational taxonomic units share limited sequence similarity with formally described taxa, suggesting that most marine diversity remains undiscovered.

Marine Microsporidia are also increasingly relevant from veterinary and public-health perspectives. *Enterocytozoon hepatopenaei* has emerged as a major constraint in penaeid shrimp aquaculture, whereas

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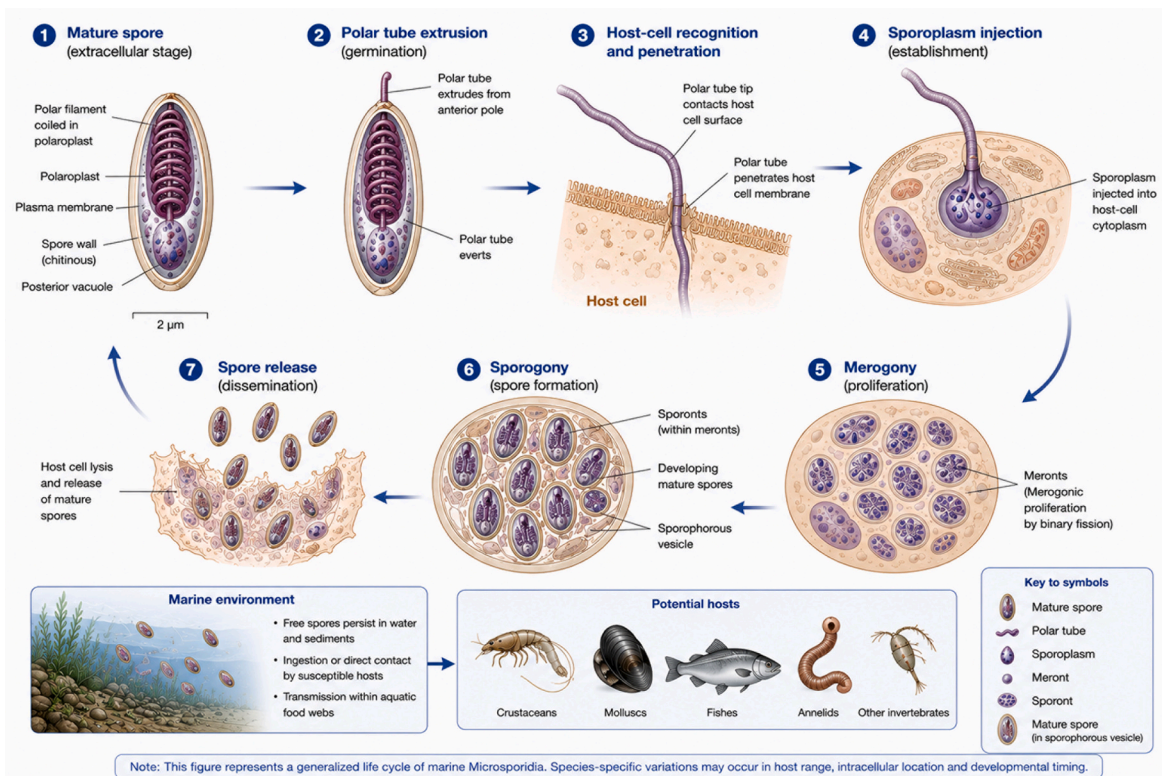


Fig. 1. Generalized life cycle and polar tube-mediated infection mechanism of marine microsporidia.

Loma salmonae, *Nucleospora salmonis* and *Desmozoon lepeophtherii* are associated with economically important diseases of salmonids [14–16]. In parallel, phylogenetic analyses have demonstrated close evolutionary relationships between human-pathogenic *Enterocytozoon bieneusi* and aquatic members of the Enterocytozoonidae [17]. Opportunistic infections caused by *Anncalia algerae*, *Trachipleistophora hominis* and *Pleistophora* spp. Further illustrate that microsporidia historically associated with aquatic or poikilothermic hosts can infect humans under suitable immunological conditions.

The present review synthesizes current knowledge regarding the biology, evolution, hidden diversity and zoonotic emergence of marine Microsporidia. Particular emphasis is placed on molecular taxonomy, environmental DNA studies, aquaculture-associated pathogens and One Health implications linking marine ecosystems, seafood production and human health.

2. Materials and methods

2.1. Literature search strategy

This work is a structured narrative (non-systematic) review; its reporting was informed by selected PRISMA 2020 principles to improve transparency, but it was not conducted as a formal systematic review, and no formal risk-of-bias assessment, quality-scoring framework or quantitative meta-analysis was applied [18]. Literature searches were performed using PubMed/MEDLINE, Web of Science Core Collection, Scopus and Google Scholar databases. Searches covered studies published between January 1975 and January 2026.

Search combinations included terms associated with Microsporidia, marine ecosystems, zoonotic transmission and aquaculture. Representative search terms included “Microsporidia”, “marine microsporidia”, “Enterocytozoonidae”, “aquaculture pathogens”, “fish microsporidia”, “environmental DNA”, “zoonosis”, “*Enterocytozoon*”, “*Nucleospora*”, “*Loma*”, “*Glugea*” and “*Desmozoon*”.

To improve transparency and partial reproducibility, representative Boolean search strings included: (“Microsporidia” OR “microsporidian”) AND (“marine” OR “aquatic” OR “aquaculture” OR “fish” OR “crustacean” OR “shrimp” OR “bivalve”); (“Enterocytozoonidae” OR “*Enterocytozoon*” OR “*Nucleospora*” OR “*Desmozoon*” OR “*Hepatospora*”) AND (“zoono” OR “One Health” OR “public health”); and (“environmental DNA” OR “eDNA” OR “metabarcoding”) AND “microsporidia”. Titles and abstracts were screened for thematic relevance, and full texts were retrieved for records addressing the biology, taxonomy, host–pathogen interactions, ecology or zoonotic potential of marine or aquatic Microsporidia; reference lists of key reviews were additionally hand-searched. Because the objective was thematic synthesis rather than quantitative pooling, records were selected purposively and no numerical PRISMA-style screening flow diagram is reported.

2.2. Inclusion criteria

Included records consisted of peer-reviewed original articles, review papers, molecular phylogenetic studies, environmental DNA surveys and clinical reports published in SCI or SCI-Expanded indexed journals. Studies focusing exclusively on terrestrial microsporidia without ecological or phylogenetic relevance to marine systems were excluded.

2.3. Data synthesis

A total of 165 records met the predefined inclusion criteria and were included in the final thematic synthesis. Due to substantial heterogeneity among host taxa, diagnostic methods and study designs, a statistical meta-analysis was not considered appropriate. Therefore, findings were synthesized narratively and thematically under sections focusing on biology, evolution, marine diversity, zoonotic emergence and One Health implications.

3. Biology and evolutionary background

3.1. Life cycle and infection apparatus

Microsporidia generally share a fundamentally conserved intracellular life cycle initiated by spore germination. Environmental or host-derived stimuli trigger rapid extrusion of the polar tube, allowing injection of sporoplasm into host cells [19]. Following invasion, parasites proliferate through merogonic and sporogonic developmental stages before mature spores are released into the environment.

The polar tube represents one of the most distinctive invasion mechanisms among eukaryotic parasites. Polar tube proteins (PTPs) mediate host-cell recognition and penetration, whereas spore wall proteins contribute to environmental persistence and transmission efficiency [19,20]. The polar tube is constructed from a small, lineage-specific protein family (Polar Tube Proteins, PTPs); to date PTP1–PTP6 have been characterised across genera such as *Encephalitozoon*, *Anncaliia* and *Nosema*, with several PTPs identified as host-cell binding mediators [4,19]. In marine systems, these adaptations facilitate transmission within dense aquatic host populations and may contribute to broad host flexibility. Spore size in marine taxa is typically 1–5 µm, although large xenoma-forming forms occur in fish and crustacean hosts [10,11,21] (Fig. 1).

3.2. Evolutionary transition from protozoa to fungi

The first molecular phylogenies of Microsporidia, based on small subunit ribosomal RNA (SSU rRNA), placed them deep at the base of the eukaryotic tree, supporting the now-rejected "Archezoa" hypothesis [22]. Subsequent phylogenetic studies based on α - and β -tubulin genes demonstrated that Microsporidia evolved from within fungal lineages [6,7]. Modern phylogenomics consistently supports their placement among highly reduced fungal-related parasites [5].

The discovery of *Mitosporidium daphniae* provided important evolutionary insight because this species retained a mitochondrial genome while already exhibiting classical microsporidian spore morphology [23]. These findings indicate that the polar-tube-based infection system evolved before the extreme genome reduction characteristic of modern Microsporidia.

3.3. Genome reduction and metabolic dependence

Microsporidian genomes are among the smallest known eukaryotic genomes. The 2.9-Mb genome of *Encephalitozoon cuniculi*, published in 2001, was the first to reveal the loss of mitochondrial genome, oxidative phosphorylation pathway, and many biosynthetic capacities, with the parasite relying on ATP imported from the host through nucleotide transporters of bacterial origin [24,25]. Comparative genomics across canonical Microsporidia has confirmed concomitant gene loss and gene-family expansion, including duplications of polar tube proteins, spore wall proteins and nucleotide transporters [26,27]. Mitochondria persist only as mitosomes, small, genome-less organelles that retain Fe-S cluster biosynthesis [28]. Members of the Enterocytozoonidae display particularly extensive metabolic reduction, reflecting their highly specialized intracellular lifestyles [17].

Comparative genomics has demonstrated that microsporidian evolution involves not only genome reduction but also lineage-specific adaptation associated with intracellular parasitism. The genome of *Encephalitozoon cuniculi* revealed extreme genome compaction, including reduced intergenic regions, loss of numerous biosynthetic pathways and strong metabolic dependence on host cells [25]. Subsequent studies confirmed that reductive evolution is a hallmark of Microsporidia, although selective retention and duplication of invasion-related genes may occur in host-adapted lineages [26,27].

A major evolutionary innovation in Microsporidia is the acquisition of nucleotide transporters that enable direct import of host-derived ATP

and other nucleotides. These transporters compensate for the loss of endogenous energy-producing pathways and represent a key mechanism underlying obligate intracellular parasitism [24,29]. Genes associated with host invasion and persistence, including polar tube proteins and spore wall proteins, are frequently retained despite extensive genome reduction.

Members of the Enterocytozoonidae, including *Enterocytozoon hepatopenaei*, *Nucleospora* spp. and *Enterocytozoon bienersi*, exhibit particularly advanced metabolic reduction and strong dependence on host-cell resources [17,30]. The close phylogenetic relationship between aquatic Enterocytozoonidae and human-associated *E. bienersi* suggests that comparative genomics may provide important insights into host switching, zoonotic emergence and cross-environmental adaptation in marine Microsporidia.

3.4. Molecular taxonomy and phylogeny

Traditional microsporidian taxonomy relied primarily on ultrastructural characters including spore morphology, polar filament coil number and sporogonic organization. However, many of these characters exhibit convergence and developmental plasticity. Molecular phylogenetics revolutionized microsporidian systematics and revealed strong ecological structuring among aquatic and terrestrial lineages [31]. Vossbrinck and Debrunner-Vossbrinck proposed a habitat-based classification recognising three classes; Aquasporidia, Marinosporidia and Terresporidia, reflecting the strong phylogenetic clustering of microsporidia infecting marine, freshwater and terrestrial hosts, respectively [31]. Subsequent re-analyses with expanded taxon sampling recovered five major clades (1–5), with Clade 5 corresponding to Marinosporidia and Clades 1 and 3 to Aquasporidia [32,33]. Marine-associated taxa are frequently grouped within Marinosporidia-related clades including fish- and crustacean-infecting genera such as *Glugea*, *Loma* and *Spraguea*. Nevertheless, environmental DNA studies demonstrate that habitat-associated phylogenetic boundaries are more permeable than previously assumed [13]. Multiple lineages appear capable of repeated transitions among marine, freshwater and terrestrial systems.

3.5. Host–pathogen molecular interactions, virulence determinants and host immune responses

At the molecular level, marine microsporidian infection is governed by a compact but highly specialized repertoire of invasion and host-manipulation factors. The polar tube proteins (PTP1–PTP6) mediate host-cell recognition and mechanical penetration, whereas spore-wall proteins govern adhesion to host surfaces and environmental persistence [4,19,20]. Because these lineages have lost most core biosynthetic and energy-generating pathways, the nucleotide transporters of bacterial origin that import host ATP and other nucleotides act as bona fide virulence determinants, coupling parasite replication directly to host-cell metabolism [24,29]. Comparative and functional studies increasingly show that microsporidia also deploy secreted effector proteins that remodel host-cell physiology from within [34], including engagement of host mitochondria through voltage-dependent anion channels by the sporoplasm-surface protein SSP1 [35] and modulation of host apoptotic signalling, as demonstrated for *Encephalitozoon*-infected cells [36].

These molecular strategies translate into reproducible patterns of tissue tropism that are both diagnostically and pathologically informative. *Enterocytozoon hepatopenaei* is restricted to the hepatopancreatic epithelium of penaeid shrimp; *Nucleospora* spp. are intranuclear parasites of haematopoietic and enteric tissues; *Loma salmonae* forms branchial xenomas following a macrophage-borne migration to the gill; and *Steinhausia* spp. target bivalve oocytes [15,21,37,38]. Xenoma-forming taxa (*Glugea*, *Loma*, *Spraguea*) sequester the parasite within a hypertrophic host cell, whereas intranuclear, non-xenomatous taxa (*Nucleospora*,

Table 1
Representative valid marine and aquatic microsporidian taxa, their verified hosts, principal tissue tropism, main diagnostic methods and associated pathological phenotypes.

Taxon	Family/clade	Principal host(s)	Tissue tropism	Diagnostics	Pathology/phenotype	Ref.
<i>Enterocytozoon hepatopenaei</i> (EHP)	Enterocytozoonidae	Penaeid shrimp (<i>Penaeus monodon</i> , <i>P. vannamei</i>)	Hepatopancreatic epithelium	Histology, ISH, nested/qPCR, RPA-CRISPR	Growth retardation, size heterogeneity; co-factor in WFS/AHPND	[16, 39, 40]
<i>Nucleospora salmonis</i>	Enterocytozoonidae	Salmonids	Intranuclear, haematopoietic	Histology, PCR	Chronic leukaemia-like disease	[38, 41]
<i>Desmozoon lepeophtherii</i> (syn. <i>Paranucleospora theridion</i>)	Enterocytozoonidae	Salmon louse <i>Lepeophtheirus salmonis</i> ; Atlantic salmon	Multi-tissue (trophic transfer)	Histology, PCR	Complex gill disease	[14]
<i>Hepatospora eriocheir</i>	Enterocytozoonidae	Crabs (<i>Eriocheir sinensis</i> , <i>Cancer pagurus</i>)	Hepatopancreas	Histology, PCR	Generalist hepatopancreatic infection	[37, 42]
<i>Enterospora canceri</i>	Enterocytozoonidae	<i>Cancer pagurus</i>	Intranuclear (hepatopancreas)	Histology, PCR	Intranuclear microsporidiosis	[37]
<i>Loma salmonae</i>	Glugeidae (Marinosporidia)	Salmonids (<i>Oncorhynchus</i> spp.)	Gill (xenomas)	Histology, PCR	Microsporidial gill disease (MGDS)	[15, 43]
<i>Glugea</i> spp.	Glugeidae (Marinosporidia)	Marine teleosts	Xenomas (visceral/subcutaneous)	Histology	Xenoma formation	[21]
<i>Spraguea</i> spp.	Marinosporidia	Anglerfish	Nervous-tissue xenomas	Histology, PCR	Xenoma disease	[21]
<i>Steinhausia mytilovum</i>	Clade IV	Mussels (<i>Mytilus</i> spp.)	Oocytes (intracytoplasmic)	Histology	Mussel egg disease; gonadal atresia	[44, 45]
<i>Myospora metanephrops</i>	Myosporidae (Crustacea)	<i>Metanephrops</i> lobsters	Skeletal muscle	Histology, PCR	Muscle microsporidiosis	[46]

Enterospora) achieve a higher degree of concealment from host immune surveillance (Table 1).

Host responses to marine Microsporidia combine innate and adaptive components. In salmonids, xenoma rupture triggers a pronounced granulomatous and inflammatory reaction, and cell-mediated immunity is central both to parasite clearance and to the strong, durable protection observed upon re-exposure; these observations underpin experimental whole-spore vaccination against microsporidial gill disease [43,47]. Notably, intranuclear and intracytoplasmic taxa that do not form xenomas appear better able to evade immune recognition, and the effector and cytokine circuitry underlying *anti*-microsporidian immunity in aquatic hosts remains only partially defined [47]. Elucidating these host–pathogen molecular interactions–invasion, effector-mediated subversion, tissue tropism and immune evasion–is central to understanding the pathogenesis, and ultimately the zoonotic potential, of this group.

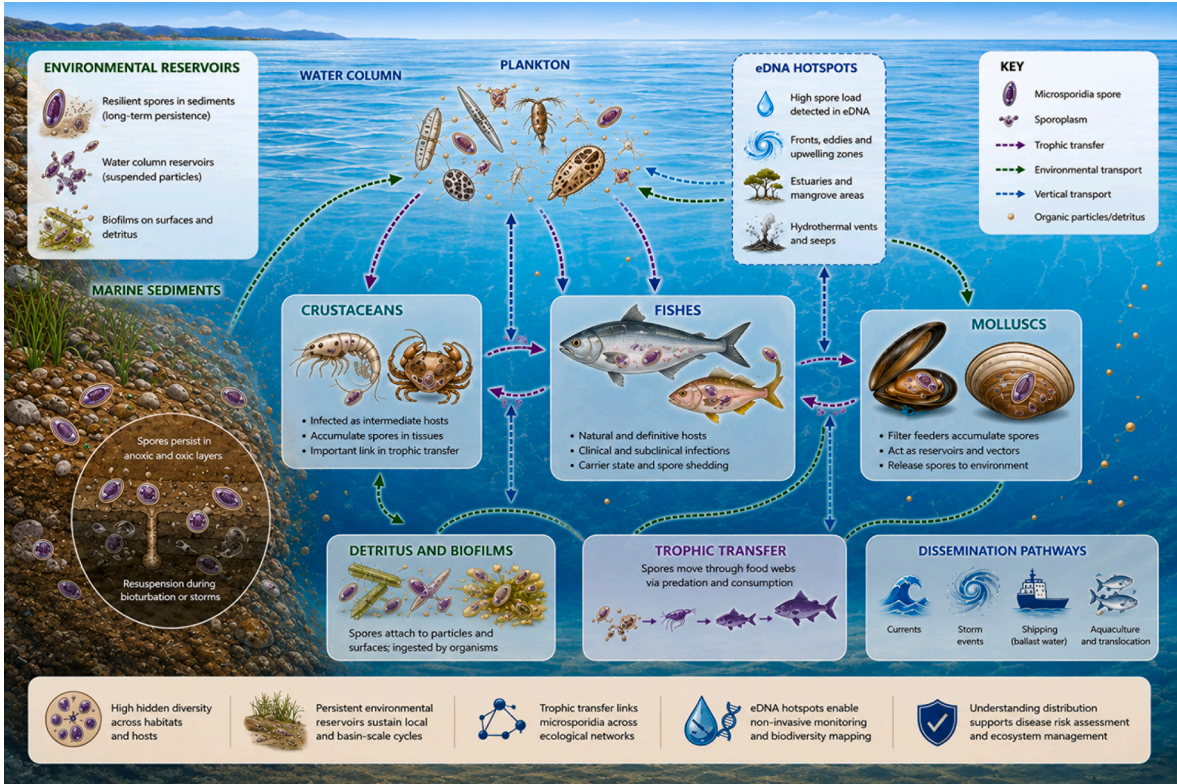


Fig. 2. Hidden diversity and ecological distribution of marine microsporidia.

4. Hidden diversity of marine microsporidia

4.1. Marine phylogenetic diversity

Marine ecosystems harbor a remarkable diversity of microsporidian parasites infecting fishes, crustaceans, copepods, molluscs and protists. Molecular analyses indicate that marine lineages occupy multiple independent phylogenetic positions rather than forming a single monophyletic radiation. Genera including *Glugea*, *Loma*, *Spraguea*, *Nucleospora*, *Hepatospora* and *Desmozoon* dominate current marine literature [21,37]. Several recently described taxa infect commercially important hosts and exhibit complex life cycles involving trophic transmission and multiple hosts. SSU phylogenies clearly separate marine clades from those of freshwater fish and from the Enterocytozoonidae infecting humans, although the latter family bridges multiple environments and is the principal subject of zoonotic concern. A new family-level taxon, Crustaceacida, has been proposed for marine lobster-infecting *Myospora metanephrops*, further expanding the class [46].

Environmental sequencing further demonstrates that described marine species likely represent only a small fraction of total diversity. Numerous operational taxonomic units identified through metabarcoding share less than 85% sequence similarity with known reference sequences [12,13]. Importantly, several environmental sequencing studies show that the previously assumed strict environmental segregation among microsporidian clades does not always hold; sequences attributed to 'marine' clades have been recovered from freshwater plankton communities, and vice versa, indicating substantial historical habitat transitions [12,13] (Fig. 2).

4.2. Crustacean-associated marine microsporidia

Crustaceans constitute one of the most economically important reservoirs of marine microsporidian infections. Stentiford and colleagues highlighted that diseases has constrained or is projected to constrain expansion of global crustacean aquaculture, with microsporidians among the principal aetiological agents [11,48]. *Enterocytozoon hepatopenaei* (EHP) is currently recognized as one of the most significant pathogens affecting penaeid shrimp aquaculture worldwide [16]. *E. hepatopenaei* (EHP), a member of the Enterocytozoonidae first described from the black tiger shrimp *Penaeus monodon* [16]. EHP infects hepatopancreatic epithelial cells and primarily causes growth retardation rather than acute mortality. Nevertheless, chronic infections result in severe economic losses due to reduced biomass production and size heterogeneity within shrimp populations [39,49]. EHP has subsequently been reported across South-East Asia, India, China and Latin America, and is increasingly recognized as a co-factor in white feces syndrome and acute hepatopancreatic necrosis disease (AHPND) of *Penaeus vannamei* [39,49,50].

Beyond shrimp, marine and estuarine crabs host a structurally diverse microsporidian fauna. *Hepatospora eriocheir* is a generalist hepatopancreatic parasite recovered from *Cancer pagurus* in the marine environment, the invasive Chinese mitten crab *Eriocheir sinensis* across estuarine and freshwater systems, and the mussel-symbiont pea crab *Pinnotheres pisum* [37]. *Enterosporea canceri*, an intranuclear parasite of *C. pagurus*, together with *Enterosporea* spp. of hermit crabs, comprise the only known intranuclear microsporidia in invertebrates [37,42]. *Myospora metanephrops* from *Metanephrops lobsters* [46], *Hyperspora aquatica*, a microsporidian hyperparasite of the paramyxian *Marteilia cochillia* infecting cockles [51], and a range of *Thelohania*-like and *Ameson*-like taxa parasitising shrimp musculature further demonstrate the ecological reach of microsporidia within marine crustacean hosts [10,11,48].

4.3. Fish-infecting marine microsporidia

microsporidian genera and more than 160 species microsporidia including economically important diseases. For instance, marine and brackish-water taxa include *Glugea* (xenoma-forming parasites of teleosts), *Loma* (gill and visceral parasites of salmonids), *Spraguea* (parasites of anglerfish), *Pleistophora* (skeletal muscle parasites of freshwater and marine fishes), *Microsporidium* (a genus for unclassified taxa), and the intranuclear *Nucleospora* [20,21]. Multi-gene phylogenies have repeatedly shown that fish-infecting microsporidia form several distinct clades, with *Glugea/Loma/Spraguea* grouping together in Marinosporida [20,21]. Fish-associated microsporidia are increasingly recognized as ecologically important parasites capable of influencing host growth, reproduction and survival in both cultured and wild populations.

Among diseases of farmed marine and anadromous fishes, microsporidial gill disease of salmon (MGDS), caused by *Loma salmonae*, produces severe branchial xenomas, secondary infections and substantial mortality in net-pen-reared *Oncorhynchus* spp. [15,43]. *Nucleospora salmonis* is an intranuclear haematopoietic parasite causing chronic leukaemia-like disease in a broad range of salmonids [38,41]. *Desmozoon lepeophtherii* (syn. *Paranucleospora theridion*) is unusual in completing parts of its life cycle in the marine ectoparasitic salmon louse *Lepeophtheirus salmonis*, with secondary infections in Atlantic salmon *Salmo salar* implicated in complex gill disease and represents a clear example of microsporidian trophic transfer across distantly related marine hosts [14]. Trophic transfer has also been documented for *Inodiosporus octospora* and *Ovipleistophora arlo* between fish and crustaceans in the marine environment, demonstrating that traditional host categories may misrepresent ecological reality [52].

4.4. Molluscan and bivalve infections

Bivalve molluscs host a small but biologically distinctive set of microsporidia. The genus *Steinhausia* comprises intracytoplasmic and occasionally intranuclear parasites of bivalve oocytes; *S. mytilovum* causes mussel egg disease in *Mytilus galloprovincialis* and *M. edulis* populations of Europe, North America, Asia and South America, with prevalences of up to 100% reported in Mediterranean farms [44,45,53]. *S. ovicola* infects *Ostrea edulis* oocytes, and *Steinhausia*-like organisms have been reported from *Saccostrea oysters* and *Ruditapes* clams [44,53]. Molecular data places *S. mytilovum* within Clade IV (Terresporidia in older usage), again indicating that habitat-of-host and phylogenetic placement do not always co-segregate [45]. Pathologically, *Steinhausia* infections trigger pronounced haemocytic inflammation, gonadal atresia and follicular disorganisation, raising concerns for mussel aquaculture and natural population dynamics [44,45].

4.5. Environmental DNA and cryptic diversity

Environmental DNA metabarcoding has transformed understanding of marine microsporidian diversity. Traditional morphology-based surveys underestimated diversity because many microsporidian species infect microscopic hosts or produce inconspicuous infections.

Group-specific sequencing approaches reveal highly divergent lineages distributed across marine sediments, planktonic communities and coastal ecosystems [13]. Many environmental sequences lack close relatives among formally described taxa, indicating the existence of extensive undiscovered diversity. For comparison, freshwater studies have demonstrated the broader applicability of microsporidian metabarcoding: in lake ecosystems, phylogenetic frameworks designed for microsporidian metabarcoding have identified thousands of OTUs distributed across seven clades, dominated by lineages whose hosts are inferred to be microeukaryotes and zooplankton rather than visible animals [12]. In marine and coastal environments, eDNA surveys have likewise reported that microsporidians yielded the largest fraction of parasite operational taxonomic units among targeted protistan groups.

Metabarcoding studies demonstrate that marine Microsporidia are not restricted to visible animal hosts. Instead, protists and zooplankton

Table 2

Evidence for the zoonotic potential of marine and aquatic Microsporidia, ranked by directness of evidence and annotated by exposure pathway (direct vs. indirect).

Evidence tier	Representative observation	Taxa	Exposure pathway	Ref.
Tier 1 – Documented human clinical case	Fatal disseminated myositis from a mosquito parasite; myositis from a genus previously known only from fish/invertebrates	<i>Anncaliia algerae</i> ; <i>Pleistophora ronneafiei</i>	Direct (opportunistic infection, immunocompromised host)	[57–59]
Tier 2 – Genotype sharing across hosts	Zoonotic ITS genotypes (D, EbpC, EbpA, Type IV) shared human/animal; 70.6% zoonotic genotypes in asymptomatic pigs	<i>Enterocytozoon bienersi</i>	Indirect (shared reservoirs; faecal–oral/waterborne)	[60–63]
Tier 3 – Environmental/eDNA detection	Human-infective <i>E. intestinalis</i> /hellem + Enterocytozoonidae in gut of edible marine fish; spores in wastewater/surface water	<i>Encephalitozoon</i> spp.; Enterocytozoonidae	Indirect (potential seafood/water vector; not demonstrated)	[64–66]
Tier 4 – Phylogenetic/mechanistic inference	<i>E. bienersi</i> nested among aquatic Enterocytozoonidae (Nucleospora); genome-reduction “ultimate opportunist” trajectory	Enterocytozoonidae (broad)	Hypothetical (no transmission demonstrated)	[17,60,67]

likely represent major reservoirs supporting hidden transmission networks within aquatic food webs. Importantly, general eukaryotic SSU primers systematically fail to amplify many canonical “long-branch” microsporidian hosts [54,55]. Interpreting these community-level signals depends critically on curated, phylogenetically ranked reference databases, because “universal” eukaryotic primers systematically under-detect long-branch microsporidian lineages [12,13,56]. Accordingly, marine-specific metabarcoding evidence should be interpreted separately from freshwater observations, with the latter serving primarily as comparative evidence for understanding host associations, microbiome interactions and methodological approaches. A marine-microbiome perspective coupling microsporidian metabarcoding with co-occurring bacterial and protistan profiling therefore represents an important future direction for resolving transmission networks within marine and coastal food webs. Importantly, the evidence considered in this section is not uniformly marine in origin. Marine-specific evidence refers to molecular, eDNA, metabarcoding, or microbiome datasets obtained directly from seawater, marine sediments, marine plankton, marine invertebrates, fishes, and marine-associated aquaculture systems. Findings from freshwater lakes and other non-marine systems are considered comparative evidence and are used to inform broader patterns of microsporidian diversity, host association, microbiome interactions, and methodological development rather than as direct evidence of marine-specific ecology. This distinction is particularly relevant because environmental sequence detection alone does not establish an active infection, a specific host association, or a marine-specific life cycle.

Emerging metabarcoding surveys indicate that marine Microsporidia are embedded within complex host-associated microbiomes, while complementary freshwater studies provide broader evidence that microsporidian infections can influence bacterial, fungal, protozoan and viral community composition. Several planktonic protists and zooplankton are also recognized as previously unsuspected microsporidian hosts [54,55]. Interpreting these community-level signals depends critically on curated, phylogenetically ranked reference databases, because “universal” eukaryotic primers systematically under-detect long-branch microsporidian lineages [12,13,56]. Accordingly, marine-specific metabarcoding evidence should be interpreted separately from freshwater observations, with the latter serving primarily as comparative evidence for understanding host associations, microbiome interactions and methodological approaches. A marine-microbiome perspective coupling microsporidian metabarcoding with co-occurring bacterial and protistan profiling therefore represents an important future direction for resolving transmission networks within marine and coastal food webs. Importantly, the evidence considered in this section is not uniformly marine in origin. Marine-specific evidence refers to molecular, eDNA, metabarcoding, or microbiome datasets obtained directly from seawater, marine sediments, marine plankton, marine invertebrates, fishes, and marine-associated aquaculture systems. Findings from freshwater lakes and other non-marine systems are considered comparative evidence and are used to inform broader patterns of microsporidian diversity, host association, microbiome interactions, and methodological development rather than as direct evidence of marine-specific ecology. This distinction is particularly relevant because environmental sequence detection alone does not establish an active infection, a specific host association, or a marine-specific life cycle.

5. Zoonotic emergence and One Health significance

Because claims of zoonotic risk are easily overstated, the evidence discussed below is organized according to a deliberate hierarchy of decreasing directness: (i) documented human clinical cases with a confirmed microsporidian aetiology; (ii) genotype- or sequence-sharing of the same taxon across human and non-human hosts; (iii) environmental or eDNA detection of human-infective taxa in water, sediment or seafood; and (iv) purely phylogenetic or mechanistic inference of host-switching potential. Throughout, these tiers are distinguished explicitly, and evolutionary relatedness or environmental detection is not treated as evidence of direct transmission. The corresponding evidence levels, together with direct versus indirect exposure pathways, are summarized in Table 2.

5.1. Human-pathogenic microsporidia

Of the more than 1500 described microsporidian species, approximately 17 species in nine genera (*Anncaliia*, *Encephalitozoon*, *Enterocytozoon*, *Microsporidium*, *Nosema*, *Pleistophora*, *Trachipleistophora*, *Tubulinosema* and *Vittaforma*) cause infections in humans, with *Enterocytozoon bienersi*, *Encephalitozoon intestinalis*, *E. hellem* and *E. cuniculi* accounting for the vast majority of confirmed cases [60,67–70]. Clinical syndromes include chronic diarrhoea and malabsorption (Enterocytozoonidae), keratoconjunctivitis (*Encephalitozoon*, *Vittaforma corneae*, *Anncaliia*), disseminated disease and encephalitis (*E. cuniculi*, *T. hominis*, *T. anthrophophthera*), and myositis (*A. algerae*, *A. connori*, *A. vesicularum*, *P. ronneafiei*, *T. hominis*, *Tubulinosema* spp.) [57,58,69–71].

5.2. Enterocytozoonidae as a bridge between aquatic and human hosts

The family Enterocytozoonidae occupies a central position in discussions of zoonotic emergence because it includes both human pathogens and aquatic parasites. It contains both *Enterocytozoon bienersi*, the predominant human-infecting microsporidian, and a series of opportunistic parasites of fish, crustaceans and copepods, including *Enterocytozoon hepatopenaei*, *Nucleospora salmonis*, *N. cyclopteri*, *N. brazilensis*, *Desmozoon lepeophtherii*, *Hepatospora eriocheir*, *Enterospira canceri*, many of which are intranuclear, possess extreme genome reduction and rely heavily on host metabolic infrastructure [17,27,37]. Stentiford et al. described the *Enterocytozoon* group as “ultimate opportunists” precisely because their evolutionary trajectory of glycolysis decay, gene loss and intracellular dependence has positioned them to colonise multiple animal phyla across all major biomes, including the human gastrointestinal tract [17]. Phylogenetically, *E. bienersi* is most closely related to fish-infecting *Nucleospora*, a finding that, since its first description in 1985, has supported the hypothesis that an ultimately marine (or freshwater fish) reservoir may underlie human microsporidiosis [60,67,72].

This evolutionary relationship suggests that aquatic ecosystems may function as reservoirs or transmission corridors for zoonotic microsporidian lineages. Members of the Enterocytozoonidae exhibit extreme genome reduction and high dependence on host metabolism, characteristics associated with opportunistic host expansion.

5.3. Waterborne transmission and seafood-associated risks

Spores of *Enterocytozoon bienersi* and *Encephalitozoon* spp. have been repeatedly detected in wastewater, surface waters and recreational aquatic environments [64,65]. Microsporidian spores are environmentally resistant and may persist for extended periods in aquatic systems. Their small size complicates removal during conventional water-treatment procedures.

Marine fishes and shellfish may therefore act as mechanical or biological vectors facilitating transfer of zoonotic spores through aquatic food chains. Although direct seafood-borne transmission to humans has

not been demonstrated to date, molecular detection of zoonotic microsporidian lineages in marine fishes suggests that marine environments should be considered within One Health surveillance strategies.

E. bienersi causes roughly 90% of all human microsporidiosis cases and is now subjected to detailed molecular epidemiological surveillance using internal transcribed spacer (ITS) genotyping, which has identified more than 600 ITS genotypes across humans, livestock, wildlife and water sources [61,62,64]. Many genotypes (e.g. EbpA, D, EbpC, Type IV) are shared across humans and multiple animal reservoirs, indicating zoonotic potential, while others appear host-restricted [61,62].

Direct detection of human-pathogenic microsporidia in edible marine fishes has now been reported. A molecular survey of 251 farmed and wild marine fishes from the Western Mediterranean (Comunidad Valenciana, Spain) detected, for the first time, the zoonotic species *Encephalitozoon intestinalis*/*E. hellem* in the digestive tract of both fish groups (in that survey, 1.45% of farmed and 7.96% of wild fish), alongside members of the Enterocytozoonidae [73]. While none of the analysed muscle samples were positive in that study, the detection of zoonotic species in the gastrointestinal tract of food fishes parallels long-recognized findings for *Cryptosporidium*, *Giardia* and *Toxoplasma gondii* in shellfish, and suggests that fish may serve as vectors moving spores from contaminated coastal waters into the human food chain [66].

5.4. Microsporidia of fish and crustaceans as direct human pathogens

Two examples illustrate that microsporidia previously regarded as fish or invertebrate parasites can directly cause human disease. *Pleistophora ronneaefti*, erected by Cali and Takvorian on the basis of an archived 1984 case of severe myositis in an immunocompromised man, was the first true *Pleistophora* species confirmed from a mammalian host; the genus had until then been recognized only from fish and possibly invertebrates [57,66]. *Anncaliia algerae* (formerly *Brachiola algerae*) is a long-established mosquito parasite, but multiple cases of fatal disseminated myositis have been documented in immunocompromised patients including a lung-transplant recipient and a woman with rheumatoid arthritis under infliximab therapy [58,59]. *Trachipleistophora hominis* and *T. anthropophthera*, originally described from AIDS patients but increasingly reported in non-AIDS contexts [69,71], group phylogenetically with parasites of cold-blooded vertebrates, again emphasising the cross-environmental potential of the phylum. The same logic applies to molecular detections in the marine domain: SSU rDNA sequences from the gastrointestinal tract of European marine fishes have been placed within the family Perezidae (Glugeida) and within the Enterocytozoonidae [73], extending the known link between marine clades and zoonotically relevant lineages.

5.5. Genotype-level evidence of zoonotic transmission

ITS-based genotyping of *E. bienersi* has provided the strongest molecular framework for inferring zoonotic transmission. Genotypes in Group 1 (e.g. D, EbpC, Type IV, EbpA) are widely distributed across humans and multiple mammalian hosts including pigs, cattle, dogs, cats and wildlife, and are often classified as zoonotic [61,62]. In a study of asymptomatic pigs, 70.6% of *E. bienersi* isolates belonged to zoonotic genotypes [63]. Detection of identical or near-identical genotypes in raw wastewater following rainfall events linked livestock fecal contamination to waterborne dispersal [65]. Although direct sampling of marine seafood for genotyped *E. bienersi* is comparatively scarce, the convergence between (i) detection of human-infectious *Encephalitozoon* and Enterocytozoonidae sequences in edible marine fishes [73], (ii) demonstration of *E. bienersi* survival in water [64,65,74,75], and (iii) the phylogenetic embedding of *E. bienersi* within a family otherwise dominated by aquatic taxa [17] supports the working hypothesis—not yet confirmed by direct transmission studies—that marine and coastal water systems may function as both transmission corridors and

reservoirs.

6. Environmental drivers of emergence

6.1. Aquaculture intensification

Industrial aquaculture creates ecological conditions highly favorable for parasite emergence. High host density reduced genetic variability and constant water exchange facilitate rapid transmission of microsporidian spores [48]. EHP outbreaks in Asian penaeid aquaculture caused production losses estimated at tens of thousands of US dollars per hectare per cycle in some surveys [39,49], while *Loma salmonae* and *Desmozoon lepeophtherii* outbreaks have caused recurrent mortalities in Pacific and Atlantic salmonid sea-pen culture [14,15,41]. Aquaculture also acts as an evolutionary amplifier because large populations of immunologically similar hosts allow rare opportunist lineages to expand, while shared water bodies bring wild and farmed populations into contact, creating opportunities for cross-environmental transfer of marine Microsporidia.

6.2. Climate change and ocean warming

Climate-driven changes in sea-surface temperature, salinity, stratification and acidification have been associated with shifts in marine microbial communities, including pathogens [76]. Although microsporidian-specific epidemiological data under climate scenarios remain limited, Stentiford and colleagues argued that mass rearing of aquatic hosts combined with global climate change is likely to exacerbate disease frequency and to facilitate more efficient transmission of opportunistic parasites in stressed or immunologically depressed hosts [11]. In the Mediterranean Sea, the documented increase in marine mucilage events associated with surface warming has been shown to harbour distinct microbial assemblages including pathogenic species absent from surrounding seawater, raising the possibility of microsporidian persistence and dispersal via such aggregates [77]. Climate-driven range expansions of invasive hosts (e.g. *Eriocheir sinensis*), shifts in migratory routes of waterfowl that carry *E. cuniculi* and *E. bienersi* spores [78,79], and altered runoff regimes that mobilise terrestrial spores into coastal systems collectively suggest that environmental change may increase opportunities for the emergence and dissemination of marine microsporidia.

6.3. Diagnostic limitations

Microsporidia remain chronically underdiagnosed in both clinical and marine contexts because spores are small and often difficult to distinguish histologically. Their spores are small, stain weakly with standard hematoxylin and eosin, and are often missed in routine histopathology unless specific stains (modified trichrome, calcofluor, Warthin–Starry) or PCR assays are applied [67,80]. In immunocompetent humans, latent and asymptomatic infections by *E. bienersi* and *Encephalitozoon* spp. are likely widespread but rarely sought [81,82]. In the marine environment, the absence of long-term parasitological monitoring programmes makes it difficult to establish whether documented increases in microsporidian prevalence reflect true emergence or improved detection [17]. Molecular epidemiology, particularly ITS-typing of *E. bienersi*, multi-locus genotyping of *Encephalitozoon* spp., and microsporidian-specific metabarcoding of marine sediments and coastal waters, represents a feasible technological foundation for integrated One Health surveillance [12,13,61].

Beyond conventional PCR and ITS genotyping, several next-generation diagnostic platforms are now maturing for microsporidian surveillance. CRISPR/Cas12a fluorescent-cleavage assays coupled with isothermal recombinase polymerase amplification enable rapid, field-deployable, single-copy-sensitive detection of aquaculture pathogens such as *Enterocytozoon hepatopenaei*, and analogous assays are being

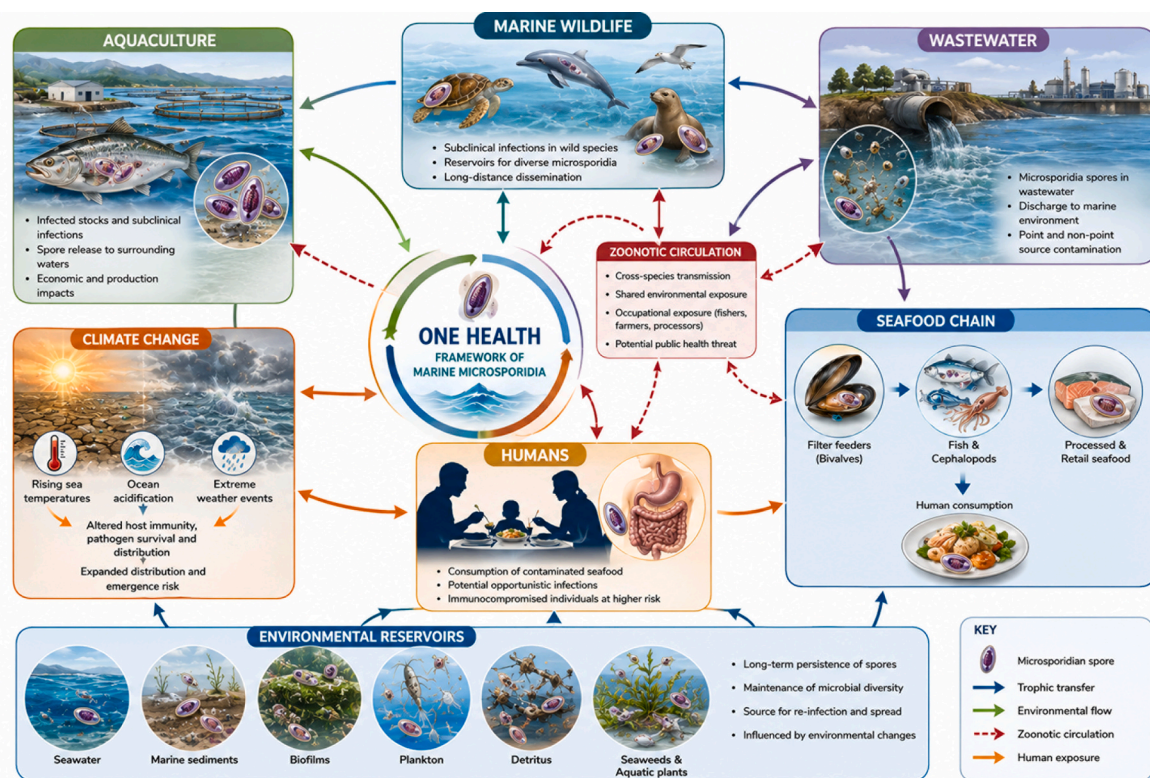


Fig. 3. One health framework of marine microsporidia.

extended to the zoonotic *Enterocytozoon bieneusi* [40]. In parallel, microsporidia-specific metabarcoding primers and curated bioinformatic reference frameworks developed across aquatic and clinical systems now permit parallel identification of both parasite and host from complex environmental or clinical DNA extracts [56,83]. For marine applications, these approaches can be combined with long-read Oxford Nanopore sequencing to provide real-time, portable detection and improved phylogenetic resolution of divergent, long-branch microsporidian lineages. Together with the microbiome-aware analyses discussed above, these tools provide a realistic technological basis for integrated One Health microsporidian surveillance.

7. One Health perspectives

Marine Microsporidia represent an emerging intersection among veterinary pathology, aquaculture sustainability, environmental health and human medicine. Their ability to infect economically important aquatic hosts while maintaining phylogenetic connections to human pathogens highlights the importance of integrated surveillance. From an ecological perspective, microsporidia influence host population dynamics, sex ratios (through feminising taxa such as *Dictyocoela* in amphipods [84], and trophic transfer dynamics between fish, crustaceans, copepods and parasitic worms [14,52]. From a public-health perspective, the convergence of (i) human pathogenic species in coastal and marine waters [64,73], (ii) phylogenetic continuity between human-infecting *E. bieneusi* and fish-infecting *Nucleospora* within the Enterocytozoonidae [17,60,67], and (iii) documented zoonotic transfer of "cold-blooded host" parasites such as *Pleistophora ronneafiei* and *Anncaliia algerae* to immunocompromised patients [36,57–59] together justify formal incorporation of microsporidia into seafood safety, water quality and immunocompromised-host risk frameworks.

A One Health response should rest on three components. First, harmonised molecular surveillance, combining microsporidian-specific metabarcoding of coastal water and sediment, sentinel screening of

farmed and wild seafood for human-pathogenic species, and clinical microsporidian diagnostics in at-risk populations, will allow detection of cross-environmental transfer in near real time [12,13,60,73]. Second, full-genome and ITS-genotype databases for both veterinary and clinical isolates are needed to support source attribution, particularly for *E. bieneusi* and *Encephalitozoon* spp. [61,62]. Third, biosecurity in aquaculture, including broodstock screening, water-supply treatment, and integrated parasite management for sea lice carrying *Desmozoon*, must be scaled up alongside human-side interventions such as immunocompromised-patient education about uncooked seafood [11, 14,17,39,48] (Fig. 3).

8. Conclusions and future perspectives

Microsporidia represent a remarkably diverse and evolutionarily dynamic group of obligate intracellular parasites whose ecological and epidemiological importance has historically been underestimated. Over the past five decades, extensive peer-reviewed research has fundamentally reshaped our understanding of this group, reclassifying Microsporidia from presumed amitochondrial "archezoa" to highly reduced fungal-related intracellular parasites [22,23]. Concurrently, nearly half of all formally described taxa have been associated with aquatic environments [10,11], and the most clinically significant human microsporidian species, *Enterocytozoon bieneusi*, has been phylogenetically linked to a broader assemblage of opportunistic marine parasites [17,60].

Recent advances in molecular phylogenetics and environmental DNA (eDNA) analyses have further revealed that marine ecosystems harbor extensive cryptic microsporidian diversity, including several lineages closely related to known human-pathogenic taxa. Nevertheless, major knowledge gaps remain unresolved. The host associations of many newly detected environmental lineages are still poorly understood [12, 13], while the genomic and metabolic mechanisms underlying cross-host adaptation within the Enterocytozoonidae remain only

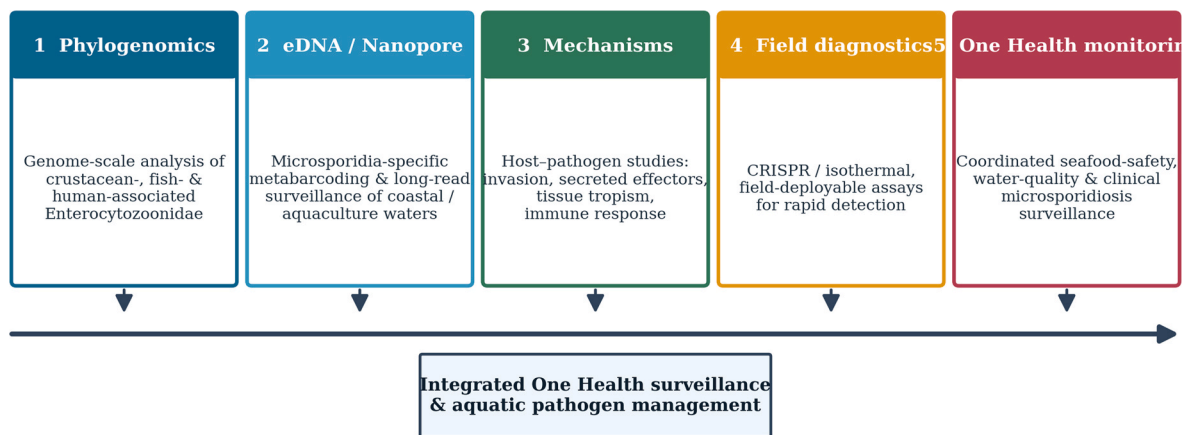


Fig. 4. Proposed future-research roadmap for marine Microsporidia, integrating: (i) genome-scale phylogenomics of crustacean-, fish- and human-associated Enterocytozoonidae; (ii) microsporidia-specific eDNA/metabarcoding and long-read (Nanopore) surveillance of coastal and aquaculture environments; (iii) mechanistic host–pathogen studies (invasion, secreted effectors, tissue tropism and host immune response); (iv) CRISPR-based, field-deployable isothermal diagnostics; and (v) coordinated One Health monitoring linking seafood safety, aquatic environmental quality and clinical microsporidiosis.

partially characterized [17,26]. Moreover, despite growing evidence for environmental circulation, comprehensive quantitative risk assessments connecting marine reservoirs with documented cases of human microsporidiosis remain limited, with only a small number of targeted genotyping studies currently available [60–62,73]. The accelerating intensification of aquaculture, ongoing climate-driven environmental change, and the expansion of global seafood trade are all likely to increase opportunities for transmission, host switching, and pathogen emergence. Although substantial ecological and epidemiological uncertainties persist, the available evidence strongly supports the inclusion of marine Microsporidia within broader One Health surveillance frameworks and integrated aquatic pathogen management strategies.

Future progress in this field will require the integration of five major research priorities (Fig. 4): (i) genome-scale phylogenomic analyses spanning crustacean-, fish-, and human-associated members of the Enterocytozoonidae; (ii) long-term, Microsporidia-specific eDNA metabarcoding and long-read Nanopore surveillance of coastal and aquaculture waters; (iii) mechanistic host–pathogen studies addressing invasion, secreted effectors, tissue tropism, and host immune responses; (iv) development of CRISPR- and isothermal-based, field-deployable assays for rapid detection; and (v) coordinated One Health monitoring integrating seafood safety, water quality, and clinical microsporidiosis surveillance.

Marine Microsporidia can no longer be regarded as peripheral or ecologically insignificant parasites. Their extensive hidden diversity, expanding environmental distribution, and increasingly recognized zoonotic potential collectively indicate that they warrant the same level of integrated, multidisciplinary attention currently directed toward other major marine pathogens.

CRediT authorship contribution statement

Çağrı Bekircan: Conceptualization, Investigation, Methodology, Project administration, Resources, Visualization, Writing – original draft, Writing – review & editing. **Muammer Kurnaz:** Data curation, Investigation, Resources, Software, Visualization, Writing – original draft.

Ethical approval

This article is a review based entirely on previously published literature. No new experiments involving humans or animals were performed and no new primary human or animal data were generated; therefore, ethical approval was not applicable.

Declaration of using generative AI and AI-assisted technologies

No generative AI or AI-assisted technologies were used to generate the scientific content, analysis, interpretation, conclusions, figures, tables, or graphical abstract of this manuscript. Any language proof-reading or grammar checking was limited to editorial-level assistance and did not contribute to the intellectual content or scientific interpretation of the manuscript. The authors take full responsibility for the accuracy, integrity, and final content of the manuscript.

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Data availability

Data will be made available on request.

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